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Lecture 1- Introduction to AI & Machine Learning

Key points:

- Intelligence is the **ability to solve problems and adapt to situations**.
- **Artificial Intelligence (AI)** aims to create systems that mimic human cognitive functions.
- AI includes several fields such as **Machine Learning, Robotics, NLP, and Computer Vision**.
- **Machine Learning (ML)** is a subset of AI that allows systems to **learn from data instead of explicit programming**.
- ML algorithms **discover patterns and relationships in data** to make predictions or decisions.
- Learning is defined by **Experience (E), Task (T), and Performance measure (P)**.
- **Data Science** involves collecting, cleaning, analyzing, and extracting insights from data.
- **Features** are input variables used to make predictions.
- **Target/Label** is the output the model predicts.
- Machine learning models aim to **map input features to outputs**.

Types of Machine Learning

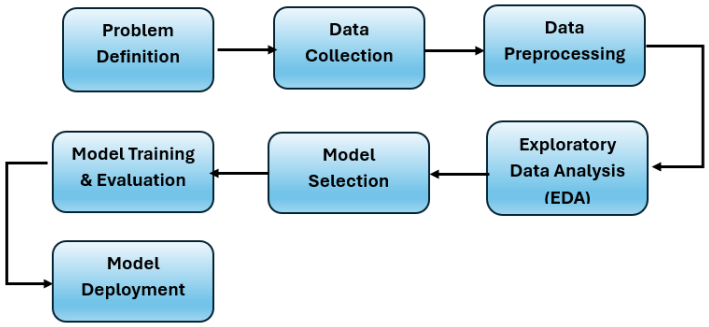
- **Supervised Learning:** uses labeled input-output data.
    - ✓ **Classification:** predicts categorical outputs (e.g., spam detection).
    - ✓ **Regression:** predicts continuous values (e.g., house price).
- **Unsupervised Learning:** works with unlabeled data to discover patterns.
    - ✓ **Clustering:** groups similar data points.
    - ✓ **Dimensionality Reduction:** reduces number of variables.
    - ✓ **Anomaly Detection:** identifies unusual data points.
    - ✓ **Association Rules:** finds relationships between items in datasets.
- **Reinforcement Learning:** agent learns through interaction with an environment using rewards.

Model Evaluation

- Used to **measure performance, compare models, and select the best algorithm**.

Lecture 2: Machine Learning Workflow & Pipeline

A machine learning project follows a **structured pipeline** consisting of several stages to transform raw data into a deployable model.



Data Preparation: Transforms raw data into a **clean and usable format**.

Includes:

- Data preprocessing
- Feature engineering

Data Preprocessing: Process of **cleaning and transforming raw data** to make it suitable for machine learning.

Common tasks:

- Data cleaning
- Handling missing values
- Handling outliers
- Data splitting
- Feature scaling
- Data transformation

Difference:

- **Noise:** random errors
- **Outliers:** extreme abnormal values

Feature Scaling: Ensures all features have **similar numeric ranges**.

Important because: Features may have different scales (e.g., age vs salary).

Common techniques: Normalization Methods & Standardization Methods

Data Transformation: Converting data into consistent formats.

Example:

- Converting categorical variables into numerical form.

Techniques:

- One-hot encoding
- Label encoding

Feature Engineering: Creating or transforming features to **improve model performance**.

Techniques:

- Feature creation: Create new features from existing)

- Feature encoding: Transform categorical variables into numeric values.
  Methods:
  - ✓ Label encoding
  - ✓ Ordinal encoding
  - ✓ One-hot encoding
  - ✓ Dummy variable encoding
  - ✓ Target encoding

- Feature binning (Discretization): Convert continuous variables into discrete categories.
  Types:

  - ✓ Equal width binning
  - ✓ Equal depth (frequency) binning
  - ✓ Quantile binning

✓ Core Idea of Lecture 2

Machine learning follows a **structured pipeline that converts raw data into a deployed predictive model through preprocessing, analysis, modeling, evaluation, and deployment.**

Lecture 3: Supervised ML (Regression Models)

Key points:

- **Supervised learning** maps input variables to output variables using labeled data.
- A supervised model represents a **mathematical function** mapping inputs $x$to outputs y.
- **Inference** is computing predictions from inputs using the trained model.
- Models contain **parameters** that control the relationship between inputs and outputs.
- **Training** means finding parameters that best predict outputs using a training dataset of input–output pairs.
- The quality of predictions is measured using a **loss function** (cost function).
- The goal of training is to **minimize the loss function**.

Types of Supervised Learning

- **Classification:** predicts discrete categories (e.g., spam detection).
- **Regression:** predicts continuous values (e.g., house price prediction).

Regression in Machine Learning

- Regression estimates the **relationship between features (inputs) and a continuous target variable**.
- Used for predictions such as:
  - House prices
  - Market trends
  - Economic forecasting
  - Correlation analysis between variables.

Types of Regression

- **Simple Linear Regression:** one independent variable predicts one dependent variable.
- **Multiple Linear Regression:** multiple independent variables predict one dependent variable.
- **Polynomial Regression:** models nonlinear relationships using polynomial functions.

Linear Regression Model

The regression equation: $y = \beta_0 + \beta_1 x + \epsilon$

- β_0: intercept
- β_1: slope (rate of change of $y$with respect to x)
- ϵ: random error term.

Residual / Prediction Error

- Residual = difference between **actual value and predicted value**.

$$e_i = y_i - \hat{y}_i$$

Least Squares Method

- Used to find the **best-fitting regression line**.
- Minimizes the **sum of squared errors (Residual Sum of Squares – RSS)**.

$$RSS = \sum(y_i - \hat{y}_i)^2$$

Model Training and Testing

- **Training:** adjust parameters to minimize loss.
- **Testing:** evaluate the model using unseen data to check **generalization ability**.

Possible problems:

- **Underfitting:** model too simple.
- **Overfitting:** model too complex.

Model Evaluation Metrics

Used to measure regression performance:

- **SST (Total Sum of Squares):** variation of actual values from their mean.
- **SSE (Sum of Squared Error):** difference between actual and predicted values.
- **SSR (Sum of Squared Regression):** variation explained by the model.

Relationship: $SST = SSR + SSE$

Important Evaluation Metrics

- **R^2 (Coefficient of Determination):** measures how well the model explains variance in the data.
- **RMSE (Root Mean Squared Error):** measures prediction error magnitude.
- **F-test:** evaluates significance of regression model.

Loss vs Cost Function

- **Loss function:** error for a single training sample.
- **Cost function:** average loss across all samples (used for optimization).

Lecture 4 – Regularization, Bias, Variance & Multicollinearity

Correlation

- **Correlation** measures the relationship between two variables.
- It indicates whether variables **move together** or **in opposite directions**.
- **Important:** Correlation **does NOT imply causation**.

Correlation Types

- **Positive correlation:** both variables increase or decrease together.
- **Negative correlation:** one increases while the other decreases.
- **No correlation:** variables have no linear relationship.

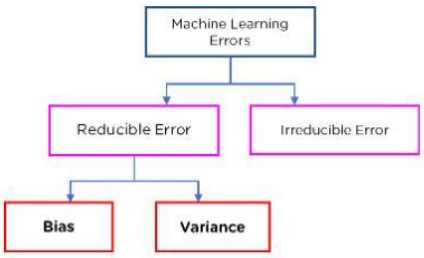
Correlation Coefficient (r)

- Measures **strength of linear relationship** between variables.
- Represented by **Pearson correlation coefficient (r)**.
- Range: **-1 to +1**.

Value of r	Meaning
+1	Perfect positive correlation
0	No correlation
-1	Perfect negative correlation

Multicollinearity: Occurs when **independent variables are highly correlated with each other**.

Machine Learning Model Error: it measures **how accurate predictions are**.



Concept	Definition	Characteristics	When High	When Low	How to Reduce
Bias	Difference between the average prediction of	Indicates how much the model assumptions simplify the true	• Model too simple	• Model fits training data well	• Use more complex models

Concept	Definition	Characteristics	When High	When Low	How to Reduce
	the model and the true value.	relationship in the data.	• Cannot capture real patterns • Leads to underfitting	• Captures underlying patterns better	• Increase number of features • Increase training data • Reduce regularization
Variance	Measures how much the model predictions change when the training data changes.	Indicates the sensitivity of the model to fluctuations in the training data.	• Model too sensitive to training data • Learns noise • Leads to overfitting	• Stable predictions • Better generalization to new data	• Cross-validation • Feature selection • Regularization • Ensemble methods • Early stopping

There is a **trade-off between bias and variance**.

Case	Result
High bias + Low variance	Underfitting
Low bias + High variance	Overfitting
Low bias + Low variance	Ideal model
High bias + High variance	Poor model

Goal: **balance both for best generalization**.

Underfitting vs Overfitting

Underfitting	Overfitting
Model too simple	Model too complex
High bias	High variance
Poor train & test accuracy	Good training but poor test performance

Regularization: it reduces **overfitting** by **adding penalties to model coefficients**.

Cost Function:

$$Cost = Loss + Regularization\ T \qquad erm$$

It forces the model to stay **simpler and more stable**.

Technique	Penalty Type	Main Idea	Key Effects	Best Used When
L1 Regularization (Lasso)	Adds absolute value of coefficients to the loss function	Forces some coefficients to shrink to exactly zero	• Performs automatic feature selection • Removes irrelevant features • Produces a simpler model	When datasets have many features and some are irrelevant
L2 Regularization (Ridge)	Adds squared value of coefficients to the loss function	Shrinks coefficients toward zero but never exactly zero	• Reduces large coefficient values • Keeps all features in the model • Helps reduce multicollinearity	When features are correlated and you want to keep all variables
Elastic Net	Combines L1 + L2 penalties	Uses both feature selection and coefficient shrinking	• Performs feature selection (L1) • Maintains stable coefficients (L2) • Balances simplicity and stability	When features are highly correlated and L1 alone removes too many features

Effect of λ (Lambda): λ controls **regularization strength ➔** Higher λ → **higher bias, lower variance**.

Lambda Value	Effect
Small λ	Model ≈ normal regression
Medium λ	Some coefficients shrink
Very large λ	All coefficients → 0

Lecture 5 - Supervised ML (Classification Part 1 logistic+KNN)

Types of Classification Problems

Type	Description	Examples
Binary Classification	Two possible classes	Spam detection, fraud detection
Multi-class Classification	More than two classes	Image classification, sentiment analysis
Multi-label Classification	One instance can belong to multiple classes	Image tagging, medical diagnosis

Logistic Regression

- A **statistical model used for binary classification**.
- Predicts the **probability that an instance belongs to a class**.

Key idea

Uses the **logistic (sigmoid) function**:

- Output values are between **0 and 1**
- Represents **probability of class membership**

Example:

- Probability patient has disease = **0.82**

Logistic vs Linear Regression

Linear Regression	Logistic Regression
Used for **continuous output**	Used for **categorical output**
Example: predicting price	Example: predicting spam/not spam
Output can be any real value	Output is probability between **0 and 1**

Maximum Likelihood Estimation (MLE)

- Used to **estimate parameters** of logistic regression.
- It finds parameters that **maximize probability of observing the data**.

Important relation: **MLE ↔ minimizing cross-entropy (log loss)**

K-Nearest Neighbors (KNN)

- A **supervised learning algorithm** that classifies data based on **nearest neighbors**.

Key assumption: **Similar data points are close to each other**

How KNN Works

1. Choose **K** (number of neighbors).
2. Calculate **distance** between new point and training data.
3. Sort distances.
4. Select **K nearest neighbors**.
5. Apply **majority voting** (classification).

For regression: Prediction = **average value of neighbors**

Choosing K in KNN

- **Small K**
  - Sensitive to noise
  - High variance
- **Large K**
  - Smooth predictions
  - Higher bias

Best K is chosen using **cross-validation or error-rate plots**.

Evaluation criteria: Confusion Matrix

A **table comparing predicted vs actual classes**.

Term	Meaning
TP	Correct positive prediction
TN	Correct negative prediction
FP	Incorrect positive prediction
FN	Incorrect negative prediction

Accuracy: it measures **how often the model predicts correctly**.

Accuracy = (Correct predictions) / (Total predictions)

Precision & Recall

Metric	Meaning
Precision	Correct positive predictions among predicted positives
Recall (Sensitivity)	Ability to find all actual positives

F1-Score

- Harmonic mean of **precision and recall**.
- Used when both metrics are important.

High F1 score = **balanced performance**.

Specificity (True Negative Rate): it measures **how well the model identifies negative cases**.

Formula concept: TN / (TN + FP)

ROC Curve & AUC

ROC Curve

- Plots:
  - **TPR (True Positive Rate)** vs
  - **FPR (False Positive Rate)**

AUC (Area Under Curve) measures classifier quality.

AUC	Meaning
1	Perfect classifier
0.5	Random guess
0	Completely wrong

Lecture 6 – Classification (SVM + Decision Trees)

Support Vector Machine (SVM)

- **SVM** is a supervised machine learning algorithm used for **classification (and sometimes regression)**.
- It works by finding the **optimal hyperplane that separates classes in the feature space**.

Important concepts

- **Hyperplane:** decision boundary separating classes.
- **Support Vectors:** data points **closest to the hyperplane** that determine the boundary.
- **Margin:** distance between the hyperplane and the nearest data points.

Goal:

- **Maximize the margin** to improve classification performance.

Linear SVM

- Works when data is **linearly separable**.
- Decision boundary formula:

$$f(x) = sign(w^T x + b)$$

Where:

- **w** → weight vector
- **b** → bias term

Soft Margin SVM

Used when data **cannot be perfectly separated**.

- Introduces **slack variables (ξ)** to allow some misclassification.
- Uses parameter **C**.

Role of C: it controls the trade-off between **Large margin and Classification error (misclassified points)**

C Value	Effect
Large C	Less tolerance for misclassification
Small C	More tolerance for errors

Non-Linear SVM

When data is **not linearly separable**, SVM maps data to a **higher-dimensional space**. This allows the model to find a **linear separator in the new space**.

Kernel Trick

A **kernel function** calculates similarity between data points and implicitly maps them into higher dimensions.

Common kernels:

- **Linear kernel**
- **Polynomial kernel**
- **RBF (Radial Basis Function)** – most common

Why RBF?

- Handles non-linear data well
- Fewer parameters
- Numerically stable

Decision Tree (DT): A **supervised learning algorithm** used for **classification and regression**. It represents decisions in a **tree-like structure**.

Structure:

Component	Meaning
Root node	Entire dataset
Internal nodes	Feature-based decisions
Branches	Outcomes of decision
Leaf nodes	Final prediction

How Decision Trees Work
Prediction process:

1. Start at the **root node**
2. Check feature condition
3. Follow the branch
4. Continue until reaching a **leaf node**
5. Output the predicted class

Splitting in Decision Trees: The algorithm splits data based on the **best attribute** to create **pure subsets** (i.e., Reduce **impurity** of data.)

Attribute Selection Measures (ASM): Two main techniques used to choose the best split:

Method	Purpose
Information Gain	Based on entropy
Gini Index	Measures impurity

Measure	Definition	Key Idea	Interpretation / Rule
Entropy	Measures the impurity or randomness in a dataset.	Indicates how mixed the classes are in a node.	• High entropy → classes are highly mixed • Low entropy → dataset is more pure • Decision trees try to reduce entropy after splitting
Information Gain	Measures the reduction in entropy after splitting the dataset using an attribute.	Shows how much information a feature gives about the class.	Formula idea: IG = Entropy(parent) − Weighted Entropy(children) Rule: choose the attribute with the highest information gain
Gini Index	Another metric that measures impurity of a node.	Used to evaluate how good a split is.	• 0 → perfectly pure node • Higher value → more impurity Best split: attribute with the lowest Gini index

Lecture 7 – Ensemble Learning

Ensemble Learning

- **Ensemble learning** combines multiple machine learning models to create a **stronger predictive model**.
- The idea is that **many weak models together perform better than a single model**.

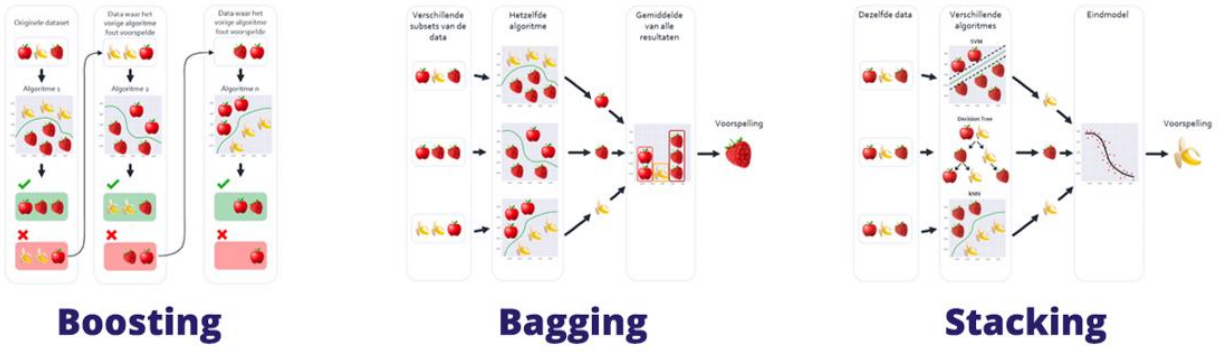
Key benefits:

- Higher **accuracy**
- Better **robustness**
- Reduced **variance and bias**

Ensemble methods combine predictions using:

- **Voting**
- **Averaging**
- **Weighted averaging**

Method	Idea
Max Voting	Class with the highest number of votes is chosen
Averaging	Average prediction of multiple models
Weighted Average	Predictions combined using different weights



Bagging (Bootstrap Aggregating)

- Ensemble technique that **reduces variance**.
- Multiple models are trained on **different bootstrap samples** of the dataset.

Key idea

- Sampling with replacement
- Train models **in parallel**

Process:

1. Create bootstrap datasets.
2. Train multiple models.
3. Combine predictions by **averaging or voting**.

Bootstrap Sampling

- Random sampling **with replacement** from the dataset.
- Some data points may appear **multiple times** in a sample.

Purpose: Create different training datasets for ensemble models.

Method	Description	Key Ideas / Steps	Result / Output
Random Forest	Ensemble model composed of many decision trees trained on different random subsets of data and features.	• Wisdom of the crowd • Feature randomness • Each tree learns from different data samples	Predictions combined using majority voting (classification) or average (regression)
Random Subspace Method	Similar to bagging but uses random subsets of features instead of samples for training models.	Steps: 1. Select random subset of features 2. Train models using those features 3. Aggregate predictions	Improves diversity between models and enhances overall performance
Extra Trees (Extremely Randomized Trees)	Ensemble method similar to Random Forest, but introduces more randomness during tree construction.	• Split values are selected randomly instead of searching for the best split	• Higher randomness • Faster training compared to Random Forest

Boosting

- Ensemble technique that **improves weak learners sequentially**.

Key idea:

- Each new model **focuses on correcting errors of previous models**.

Steps:

1. Train first model.

2. Increase weight for misclassified samples.
3. Train next model.
4. Combine models using **weighted predictions**.

Goal:

- **Reduce bias and improve accuracy**.

Method	Description	Main Idea / Process	Key Features / Result
AdaBoost (Adaptive Boosting)	Boosting algorithm that adjusts sample weights during training.	Process: 1. Train weak classifier 2. Increase weight of misclassified samples 3. Train next classifier focusing on difficult samples 4. Combine classifiers using weighted voting	Improves performance by focusing on hard-to-classify data
Gradient Boosting	Sequential ensemble algorithm that builds models to correct errors of previous models.	Steps: 1. Make initial prediction 2. Compute residual errors 3. Train new model on residuals 4. Update predictions iteratively	Gradually minimizes prediction errors and improves model accuracy
XGBoost (Extreme Gradient Boosting)	Optimized and efficient implementation of Gradient Boosting.	Builds decision trees sequentially while optimizing the loss function and correcting previous errors.	• High computational efficiency • Regularization to prevent overfitting • Handles missing values • Not sensitive to outliers • No feature scaling required

Bagging vs Boosting

Feature	Bagging	Boosting
Training	Parallel	Sequential
Data	Random sampling	Weighted sampling
Main goal	Reduce variance	Reduce bias

Feature	Bagging	Boosting
Example	Random Forest	AdaBoost

Stacking (Stacked Generalization): Ensemble technique that **combines different models using another model**.

Structure:

Level	Description
Level-0 (Base Models)	Multiple ML models trained on dataset
Level-1 (Meta Model)	Model that combines predictions

Example:

- Base models: SVM, Decision Tree, Random Forest
- Meta model: Logistic Regression

Lecture 8 – Model Evaluation, Cross-Validation & Hyperparameter Tuning.

Model Evaluation

- Measures how well a model **generalizes to unseen data**.
- Helps detect **overfitting and underfitting**.

Common metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- RMSE

Cross-validation methods

Method	Description	Process / Key Idea	Advantages	Disadvantages / Notes
Hold-Out Method	Simple model evaluation technique that splits the dataset into training and testing sets.	1. Split dataset into Training (70–80%) and Test (20–30%) 2. Train model on training data 3. Evaluate model on test data	• Simple • Fast to implement	• Results depend on random data split
Cross-Validation	Statistical method used to evaluate model performance more reliably by repeatedly splitting data.	Model is trained and tested multiple times on different subsets of data.	• Reduces bias and variance • More reliable evaluation	More computational cost than hold-out
K-Fold Cross-Validation	Dataset is divided into K equal parts (folds) and evaluated multiple times.	1. Split dataset into K folds 2. Train on K−1 folds 3. Test on remaining fold 4. Repeat K times 5. Average results	Provides stable performance estimate	Slightly higher computation
Stratified K-Fold Cross-Validation	Improved K-Fold that preserves class distribution in each fold.	Each fold keeps the same class ratio as the full dataset.	Best for imbalanced datasets	Not suitable for time-series data

Method	Description	Process / Key Idea	Advantages	Disadvantages / Notes
Leave-One-Out Cross-Validation (LOOCV)	Special case of K-Fold where K = number of samples.	• Use one sample as test • Train on n−1 samples • Repeat for all samples	• Very accurate evaluation	Computationally expensive
Repeated K-Fold Cross-Validation	Runs K-Fold multiple times with different random splits.	Example: 5-Fold repeated 3 times → 15 evaluations	• Improves reliability • Reduces variance	Higher computational cost

Hyperparameters: Parameters **set before training** the model.

Algorithm	Hyperparameters
Polynomial Regression	Degree
Logistic Regression	Regularization
Decision Tree	Max depth
Random Forest	Number of features
Neural Network	Layers, neurons, learning rate

Hyperparameter Space: The set of **all possible hyperparameter combinations**.

Properties:

- Each axis = one hyperparameter
- Each point = unique parameter combination.

Goal:

- Find the **optimal combination**.

Hyperparameter Tuning Methods

Method	Idea
Manual Search	Trial-and-error
Grid Search	Test all combinations
Random Search	Random combinations
Halving Grid Search	Eliminate weak combinations early
Halving Random Search	Random search with resource reduction
Bayesian Optimization	Predict best parameters using probability

Lecture 9 – Unsupervised Machine Learning (Clustering, PCA, Association Rules).

Unsupervised Machine Learning

- In **unsupervised learning**, the **correct labels or classes are not known**.
- The goal is to **discover hidden patterns or structures in data**.

Main tasks:

- Clustering
- Dimensionality Reduction
- Association Rule Mining

Clustering

- **Clustering** groups similar data points into **clusters based on similarity**.

Goal: **High intra-cluster similarity and Low inter-cluster similarity**

Applications:

- Customer segmentation
- Document grouping
- Image analysis
- Market segmentation

Types of Clustering Approaches

Approach	Idea
Centroid-based	Assign points to nearest cluster center
Hierarchical	Merge or split clusters progressively

Common algorithms:

- **K-Means**
- **Hierarchical clustering**

Distance Measures in Clustering: determine **similarity between points**.

Common measures:

- **Euclidean distance (L2 norm)**
- **Manhattan distance (L1 norm)**
- **Maximum distance (L∞ norm)**

K-Means Clustering

- A centroid-based clustering algorithm.
- Divides data into **K clusters**.

Goal: Minimize distance between data points and cluster centroid.

K-Means Algorithm Steps

1. Choose **K cluster centroids** randomly.
2. Assign each data point to the **nearest centroid**.
3. Recalculate centroids (mean of cluster points).
4. Repeat until **cluster assignments stop changing**.

Determining Optimal Number of Clusters

Method	Description	Key Idea	Interpretation
Elbow Method	Uses Sum of Squared Errors (SSE) to evaluate clustering performance for different values of K.	Plot K vs SSE and observe the curve. The optimal K is at the "elbow point", where the decrease in SSE starts slowing down significantly.	The elbow point indicates a balance between low error and model simplicity.
Silhouette Method	Measures how well each data point fits within its assigned cluster compared to other clusters.	Computes a Silhouette score for each point.	+1: well clustered 0: clusters overlap −1: misclassified point

Hierarchical Clustering: Groups data by **building a hierarchy of clusters**.

Type	Description
Agglomerative (Bottom-Up)	Start with individual points → merge clusters
Divisive (Top-Down)	Start with one cluster → split repeatedly

Visualization: **Dendrogram (tree diagram)**

Linkage Methods in Hierarchical Clustering: Used to measure distance between clusters.

Method	Meaning
Single linkage	Distance between closest points
Complete linkage	Distance between farthest points
Average linkage	Average distance between clusters

Dimensionality Reduction: Technique used to **reduce number of input variables (features)**.

Benefits: Faster computation, Less noise, Better visualization, and Reduced overfitting
Common method: PCA (Principal Component Analysis)

Principal Component Analysis (PCA): it transforms correlated variables into **new uncorrelated components** called **principal components**.

Goal: Preserve **maximum variance** with fewer dimensions.

PCA Concepts

Term	Meaning
Eigenvectors	Directions of maximum variance
Eigenvalues	Amount of variance explained

Rules:

- **PC1 captures most variance**
- **PC2 captures second largest variance**

PCA Steps

1. **Standardize data**
2. Compute **covariance matrix**
3. Calculate **eigenvalues and eigenvectors**
4. Select principal components
5. Transform data to new feature space

Association Rule Learning: Technique used to **discover relationships between items in large datasets**.

Example: Market Basket Analysis
Example rule: Bread → Butter
Meaning: Customers who buy bread often buy butter.

Association Rule Metrics:

Metric	Meaning
Support	Frequency of itemset in dataset
Confidence	Probability of Y when X occurs
Lift	Strength of association between items • Lift > 1 → strong association • Lift = 1 → no association

Lecture 10 – Artificial Neural Networks (ANN) and Backpropagation.

AI, ML, and Deep Learning

- **Machine Learning (ML):** Branch of AI that uses statistical algorithms to learn from data and make predictions.
- **Deep Learning (DL):** Subfield of ML that uses **artificial neural networks with multiple layers**.

Key difference:

- ML often requires **manual feature engineering**.
- DL **automatically learns features from data**.

Artificial Neural Network (ANN)

- ANN is a **computational model inspired by the human brain**.
- It consists of **artificial neurons connected by weighted links**.

Purpose: Transform **input data → output prediction**.

Artificial Neuron Concept

A neuron performs computation using:

- **Inputs (x)**
- **Weights (w)**
- **Bias (b)**
- **Activation function**

Output depends on the **weighted sum of inputs**.

General form: $z = w \cdot x + b$ ➔ Neuron "fires" when the value exceeds a **threshold**.

Weights and Bias

- **Weights:** Represent importance of each input.
- **Bias:** Helps shift the decision boundary.

Weighted sum determines neuron output.

Activation Functions: determine the **output of a neuron** and introduce **non-linearity**.

Common activation functions:

Activation Function	Purpose
Binary Step	Outputs 0 or 1
Linear	Used in regression
Sigmoid	Outputs between 0 and 1
Tanh	Outputs between -1 and 1
ReLU	Most common in deep learning
Softmax	Used for multi-class classification

Perceptron

- Simplest neural network model.
- Used for **binary classification**.

Key properties:

- Finds a **linear decision boundary**.
- Works only for **linearly separable data**.

Example tasks: AND gate classification.

Classical / Error correction Perceptron Learning Rule (Weights are updated when predictions are wrong).

Update formula:

$$w_{new} = w_{old} + \alpha \times output \times input$$

$$w_{new} = w_{old} + \alpha(desired - output) \times input$$

Where: **α = learning rate**

Typical Neural Network structure

Layer	Role
Input Layer	Receives input features
Hidden Layer(s)	Extract patterns and features
Output Layer	Produces prediction

Hidden Layers: allow neural networks to **learn complex patterns**.

Important properties:

- Detect hidden features in data
- Enable modeling of **non-linear relationships**.

With:

- 1 hidden layer → represent continuous functions
- Multiple layers → solve complex problems.

Feedforward Neural Network

- Data flows **only in one direction**:

Input → Hidden Layer → Output.

Characteristics:

- No loops
- Used in basic neural networks.

Multilayer Perceptron (MLP)

Extension of perceptron with **multiple hidden layers**.

Features:

- Fully connected layers
- Uses **gradient descent for learning**
- Supports **forward propagation and backpropagation**.

Learning in ANN

ANN learns by **adjusting weights based on training data**.

Process:

1. Initialize weights randomly.
2. Pass input through network.
3. Compute prediction error.
4. Adjust weights to minimize error.

Forward Propagation

- Input moves **forward through layers** to compute output.

Steps:

1. Compute weighted sums
2. Apply activation functions
3. Produce output prediction.

Backpropagation

Algorithm used to **update weights by propagating errors backward**.

Process:

1. Perform forward pass.
2. Compute output error.
3. Propagate error backwards through network.
4. Update weights using gradients.

Goal:

- **Minimize total network error**.

Gradient Descent

Optimization method used to **minimize loss function**.

Key idea:

- Move weights in direction that **reduces error**.

Variants:

- Batch Gradient Descent
- Stochastic Gradient Descent (SGD)
- Mini-batch Gradient Descent

Training Terminology

Term	Meaning
Batch	Subset of training data
Epoch	One full pass through entire dataset
Iteration	One forward + backward pass

Example:

- Dataset: 10,000 samples
- Batch size: 1,000
- Batches per epoch = 10.